

CAIE Course Program

Assignment - Student Grade System | learn. build. operate

1. What you will build

Build a **Streamlit** web app that asks the user to enter a mark (a number between 0 and 100) and then shows the matching letter grade. The app runs in the browser, so there is no terminal menu and no other framework. You will set up a clean Python environment, install Streamlit, write your code in a **.py** file, run it with `streamlit run`, and confirm it produces the correct output. Use **Streamlit only** - no other web framework or UI library.

2. Grading scale to implement

Your app must convert a mark into a grade using exactly this scale:

Mark range	Grade
90 – 100	A
80 – 89	B
70 – 79	C
60 – 69	D
Below 60	E

Note: the boundaries are inclusive - a mark of exactly 90 is an A, exactly 80 is a B, and so on.

3. Requirements

- Use **Streamlit only**. Do not use any other web framework or UI library (no Flask, Django, FastAPI or Gradio).
- Take the mark from the user through a Streamlit input widget in the browser.
- Show a clear message displaying the entered mark and the resulting grade.
- Handle the full 0–100 range and cover every grade band above.
- Keep all your code in a single file named `grade_system.py`.

Think about edge cases

What should happen if someone enters a number above 100, below 0, or leaves the field empty? Decide how your app should respond, and make sure it does not crash unexpectedly.

4. Step-by-step setup

Follow these steps in order. Commands are shown for both macOS/Linux and Windows where they differ.

Step 1 - Create a project folder

```
mkdir grade-system  
cd grade-system
```

Step 2 - Create a virtual environment (venv)

A virtual environment keeps this project isolated from the rest of your system.

```
python -m venv venv  
# (on some systems use 'python3' instead of 'python')
```

Step 3 - Activate the virtual environment

```
# macOS / Linux:  
source venv/bin/activate  
  
# Windows (PowerShell):  
venv\Scripts\Activate.ps1  
  
# Windows (Command Prompt):  
venv\Scripts\activate.bat
```

When activated, you will see (venv) at the start of your terminal prompt.

Step 4 - Install Streamlit

Streamlit is the only package this assignment needs.

```
pip install streamlit
```

Step 5 - Run your app

Streamlit apps are started with `streamlit run`, not `python`.

```
streamlit run file_name.py
```

The app opens in your browser at `http://localhost:8501`. Test it several times with different marks to confirm each grade band works.

Step 6 - Stop and deactivate when finished

Press `Ctrl + C` in the terminal to stop the app, then deactivate the venv.

```
deactivate
```

5. Example of how a run should look

Your layout and wording can differ, but a single run should behave roughly like this:

```
Student Grade System

Enter your mark (0-100): [ 85 ]

Mark: 85 -> Grade: B
```

6. What to submit

- Your `grade_system.py` file.
- A short screenshot or screen recording of the app in the browser showing at least three different marks.
- Two or three sentences explaining how you handled invalid input.
- Upload the code to GitHub and send us the repository link.

Reminder: this is a solo task using Streamlit only. Build the logic yourself - the steps above are your roadmap, not the answer. Good luck!